

Notes: Tang (JFE, 2009)

Information Asymmetry and Firms' Credit Market Access: Evidence from Moody's Credit Rating Format Refinement

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1 Big picture

This paper studies whether reductions in **credit-market information asymmetry** change firms' financing and real investment decisions. The empirical setting is Moody's 1982 rating format refinement. Before April 26, 1982, Moody's reported corporate credit quality using broad letter categories. On April 26, 1982, Moody's introduced numerical modifiers - 1, 2, and 3 - within several broad rating categories. The same broad rating could now be split into finer subcategories such as A1, A2, and A3.

The key identifying idea is that this refinement did not reflect a new firm-specific event. Moody's stated that the numerical modifiers were refinements of existing categories and that rating definitions and procedures remained unchanged. The paper therefore interprets the refinement as a shock to the **public precision of credit information**: the rating agency already possessed information, but the market received a more precise public signal.

Main finding

Firms that receive favorable refined ratings experience better credit-market access and subsequently change their real decisions:

- **Borrowing costs fall.** Upgraded firms experience roughly a 20 basis point lower post-refinement yield spread relative to downgraded firms.
- **Debt issuance rises.** Upgraded firms issue about 2 percent of lagged assets more long-term debt than downgraded firms, and around 5 percent more net long-term debt.
- **Financing mix shifts toward debt.** Upgraded firms issue less equity and rely more on debt financing.
- **Real investment responds.** Upgraded firms invest more, accumulate less cash, and grow assets faster than downgraded firms.

Interpretation: Credit ratings are not just labels. More precise credit information can relax credit-market frictions, lower borrowing costs for better-quality firms, and translate into real investment and growth.

2 Incremental contribution of the paper

1. **Identification contribution.** The paper uses Moody’s 1982 rating refinement as a plausibly exogenous shock to information precision. This helps avoid the usual problem that standard information-asymmetry proxies, such as firm size or analyst dispersion, are correlated with investment opportunities.
2. **Credit-market access channel.** Prior work showed that ratings affect bond and stock prices. Tang pushes further by showing that ratings affect firms’ financing quantities and investment decisions.
3. **Real effects of information asymmetry.** The paper directly links a reduction in credit-market information asymmetry to real corporate outcomes: capital expenditures, cash accumulation, and asset growth.
4. **Debt maturity and capital-structure mechanism.** The results are consistent with Diamond-style debt maturity logic: firms that had been pooled with lower-quality borrowers gain access to cheaper long-term debt once better credit quality is revealed.
5. **Robustness contribution.** The paper shows that the results survive controls for ex ante yield spreads, S&P ratings, within-rating-class refinements, industry fixed effects, and pre-refinement financing patterns.

3 Institutional background

3.1 Moody’s 1982 rating refinement

Before the refinement, Moody’s assigned firms to broad categories such as Aaa, Aa, A, Baa, Ba, B, Caa, Ca, and C. After the refinement, numerical modifiers were attached to the categories Aa, A, Baa, Ba, and B:

- Modifier 1: best credit quality within the broad rating category.
- Modifier 2: middle credit quality within the broad rating category.
- Modifier 3: worst credit quality within the broad rating category.

Thus a firm that was previously rated A could become A1, A2, or A3; firms near category boundaries could also move to nearby refined categories such as Aa3 or Baa1. The refinement expanded the effective corporate rating scale from coarse to more granular categories.

3.2 Why this setting is useful

The paper emphasizes three useful institutional features:

- **No firm-level trigger.** The refinement was not caused by changes in firms’ fundamentals.

- **Same underlying information at Moody’s.** Moody’s stated that rating definitions and procedures did not change.
- **Simultaneous implementation.** The refinement applied to all followed bonds at the same time and was not preceded by a public announcement.

Interpretation: The empirical variation comes from the public revelation of finer credit-quality information, not from a sequence of separate firm-specific credit events.

4 Conceptual framework

The mechanism is easiest to see in a two-type borrower model. Suppose good and bad borrowers are pooled in the same coarse Moody’s rating class. Before refinement, creditors cannot fully separate them, so they lend at a pooled interest rate. After refinement, the better borrowers receive favorable modifiers and the weaker borrowers receive unfavorable modifiers.

For the good firms, the refinement lowers the cost of borrowing:

$$\Delta r_i < 0 \quad \text{for firms receiving favorable refined ratings.}$$

Lower borrowing costs make previously delayed projects more attractive and increase debt capacity. Therefore, upgraded firms should subsequently issue more debt and invest more.

4.1 Debt maturity logic

The paper connects this intuition to Diamond’s debt maturity model. Firms that believe they are under-rated may delay investment or use short-term debt while waiting for their true credit quality to be revealed. When the rating refinement reveals better quality, these firms can shift toward long-term debt financing.

4.2 Predictions

1. Upgraded firms should experience a lower cost of borrowing than downgraded firms.
2. Upgraded firms should issue more long-term debt and more net debt.
3. Upgraded firms should rely less on equity financing.
4. Upgraded firms should invest more, save less cash, and grow faster.

5 Data and measurement

5.1 Data sources

The empirical analysis combines several sources:

- Moody’s senior rating database and the April 26, 1982 issue of *Moody’s Bond Record*.
- Lehman Brothers Fixed Income Database for corporate bond yields and Treasury yields.
- Compustat quarterly data for firm financial statements.
- S&P’s March 1982 *Bond Guide* for preexisting S&P ratings.

5.2 Samples

The initial Moody’s refinement sample includes 1,193 firms. Of these, 363 receive modifier 1, 391 receive modifier 2, and 439 receive modifier 3. The cost-of-borrowing sample contains 704 firms with bond yield information. The final merged sample used for financing and investment regressions contains 266 firms with Compustat and bond-yield data.

5.3 Main outcome variables

- **Cost of borrowing:** corporate bond yield spread relative to comparable Treasury yields, and alternative spread measures.
- **Financing decisions:** long-term debt issuance, short-term debt issuance, total debt issuance, net debt issuance, and equity issuance, scaled by lagged assets.
- **Real outcomes:** capital expenditures, cash accumulation, and asset growth, scaled by lagged assets.

5.4 Controls

The regressions control for firm size, market-to-book, profitability, asset tangibility, lagged financing decisions, ex ante yield spreads, and rating-class fixed effects. Robustness tests add S&P ratings and industry fixed effects.

6 Empirical design

6.1 Borrowing-cost regression

The paper first tests whether refined ratings affect subsequent yield spreads:

$$\Delta YieldSpread_{i,t} = \beta_0 + \beta_1 Upgrade_i + \beta_2 NoChange_i + \gamma_1 A_{i,t-1} + \gamma_2 Baa_{i,t-1} + \gamma_3 Ba_{i,t-1} + \varepsilon_{i,t}.$$

Downgrades are the omitted group, so β_1 measures the change in yield spreads for upgraded firms relative to downgraded firms.

6.2 Financing and investment regressions

The main firm-policy regressions use first differences:

$$\Delta Y_{i,t} = \beta_0 + \beta_1 Upgrade_i + \beta_2 NoChange_i + \beta_3 YieldSpread_{i,t-1} + \delta' \Delta X_{i,t-1} + \text{Rating FE} + \varepsilon_{i,t}.$$

Here $Y_{i,t}$ can be debt issuance, equity issuance, capital expenditures, cash accumulation, or asset growth. The first-difference design removes time-invariant firm effects and compares how policies change after the refinement relative to before.

6.3 Identification logic

The key comparison is between firms that were similar enough to be in the same coarse rating system before April 1982 but were separated by the new finer scale. The identifying assumption is

that the refinement changed the market's information about the firm rather than the firm's actual investment opportunities.

7 Tables and figures

7.1 Table 1. Moody's rating format refinement and Table 2. Summary statistics for the matched firm sample

Table 1

Moody's 1982 rating format refinement. On April 26, 1982, Moody's refined its previously coarse rating classes by adding modifiers to rating classes Aa, A, Baa, Ba, and B. Modifiers "1", "2", and "3" represent subratings of highest, average, and worst credit quality, respectively.

Credit risk	Moody's rating categories	
	Pre-refinement	Post-refinement
Investment grade	Aaa	Aaa
	Aa	Aa1
		Aa2
		Aa3
	A	A1
		A2
		A3
	Baa	Baa1
		Baa2
		Baa3
Non-investment grade	Ba	Ba1
		Ba2
		Ba3
	B	B1
		B2
		B3
	Caa	Caa
	Ca	Ca
	C	C

Table 1. Moody's rating format refinement.

Table 2

Summary statistics of key firm characteristics for the 266 firms in the matched sample. *TotalAssets* is the book value of total assets; *MB/BV* is the market-to-book ratio; *Profitability* is the operating income before depreciation and scaled by total assets; *Intangibility* is firms' net property, plant, and equipment scaled by the total assets; *Leverage* is the book value of debt scaled by total assets; *Cash* is the cash and cash equivalents scaled by total assets; *Long-Term Debts* is firms' issuance of long-term debt scaled by total assets; *Net Long-Term Debts* is firms' issuance of long-term debt net of change in cash and cash equivalents scaled by total assets; *EquityIss* is firms' equity issuance scaled by lagged total assets; *CapEx* is firms' capital expenditure, which is calculated as the change in firms' property, plant, and equipment plus depreciation and scaled by lagged total assets; *CashAccumulation* is defined as the change in firms' cash and cash equivalents and scaled by lagged total assets; and *AssetGrowth* is calculated as the change in firm's book value of total assets and scaled by lagged total assets. Time t is the time of the rating refinement. All accounting variables are yearly results calculated based on quarterly Compustat data.

N = 266	Mean	Standard deviation	Percentile		
			10th	50th	90th
<i>TotalAssets_t</i> (US\$M)	4.760	11.836	298	1,851	9,034
<i>MB/BV_t</i>	0.933	0.234	0.755	0.898	1.130
<i>Profitability_t</i>	0.123	0.067	0.019	0.123	0.196
<i>Intangibility_t</i>	0.538	0.284	0.021	0.593	0.875
<i>Leverage_t</i>	0.326	0.139	0.158	0.317	0.501
<i>Cash_t</i>	0.052	0.072	0.003	0.021	0.168
<i>Long-Term Debts_t</i>	0.045	0.100	-0.018	0.012	0.144
<i>Net Long-Term Debts_t</i>	0.046	0.106	-0.039	0.026	0.142
<i>EquityIss_t</i>	0.018	0.038	-0.001	0.005	0.050
<i>CapEx_t</i>	0.123	0.145	0.004	0.094	0.241
<i>CashAccumulation_t</i>	0.000	0.041	-0.044	-0.001	0.034
<i>AssetGrowth_t</i>	0.123	0.173	-0.005	0.086	0.259
Δ <i>TotalAssets_{t-1,t}</i> (US\$M)	448	1,257	-4	118	845
Δ <i>MB/BV_{t-1,t}</i>	-0.119	0.223	-0.328	-0.044	0.031
Δ <i>Profitability_{t-1,t}</i>	-0.003	0.030	-0.040	-0.001	0.024
Δ <i>Intangibility_{t-1,t}</i>	0.007	0.040	-0.033	0.004	0.052
Δ <i>Long-Term Debts_{t-1,t}</i>	0.015	0.099	-0.080	0.000	0.125
Δ <i>Net Long-Term Debts_{t-1,t}</i>	0.025	0.114	-0.088	0.008	0.143
Δ <i>EquityIss_{t-1,t}</i>	-0.008	0.072	-0.054	0.000	0.031
Δ <i>CapEx_{t-1,t}</i>	-0.012	0.094	-0.071	-0.005	0.064
Δ <i>CashAccumulation_{t-1,t}</i>	-0.013	0.065	-0.089	-0.003	0.044
Δ <i>AssetGrowth_{t-1,t}</i>	-0.028	0.158	-0.167	-0.020	0.095
Δ <i>Long-Term Debts_{t,t+1}</i>	-0.014	0.083	-0.112	-0.003	0.066
Δ <i>Net Long-Term Debts_{t,t+1}</i>	-0.012	0.097	-0.147	-0.019	0.058
Δ <i>EquityIss_{t,t+1}</i>	0.007	0.050	-0.027	0.001	0.043
Δ <i>CapEx_{t,t+1}</i>	-0.044	0.121	-0.164	-0.017	0.039
Δ <i>CashAccumulation_{t,t+1}</i>	0.014	0.056	-0.023	0.007	0.070
Δ <i>AssetGrowth_{t,t+1}</i>	-0.047	0.178	-0.223	-0.030	0.140

Table 2. Summary statistics for the matched firm sample.

Read:

- Table 1 shows the institutional treatment: broad ratings are split into finer subratings. A1 is better than A2, and A3 is worse than A2.
- Table 2 shows that the matched sample consists of large public firms, with mean total assets of about \$4.76 billion and mean leverage of 0.326.
- The sample firms have meaningful pre-refinement debt issuance and investment, making it possible to study subsequent financing and real responses.

Economic message: Table 1 clarifies why the refinement changes information precision without necessarily changing fundamentals. Table 2 shows that the sample consists of firms large enough to access public debt markets, where credit ratings should matter for financing costs and debt capacity.

7.2 Figure 1. Financing decisions around the refinement and Figure 2. Investment and cash decisions around the refinement

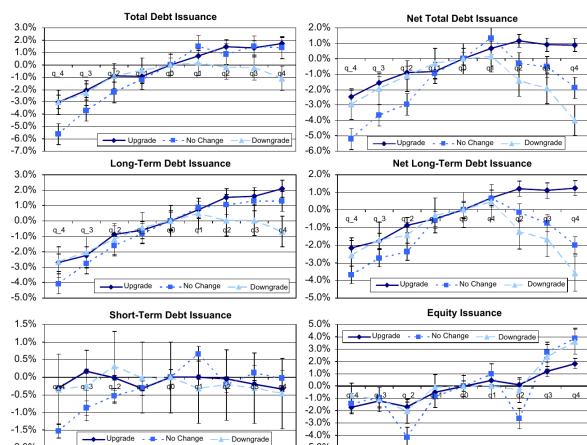


Fig. 1. Effects of rating refinement changes on firm's financing decisions. The graphs show firms' cumulative financing decisions in the four quarters pre- and post-Moody's 1982 rating refinement (denoted as quarter "0" in the graphs). All graphs are re-centered at quarter 0. *Total Debt Issuance* is defined as the issuance of long-term debt plus short-term debt; *Net Total Debt Issuance* is defined as total debt issuance net of the change in cash and cash equivalents; *Long-Term Debt Issuance* is defined as the issuance of long-term debt; *Net Long-Term Debt Issuance* is defined as the issuance of long-term debt net of the change in cash and cash equivalents; *Short-Term Debt Issuance* is defined as the issuance of short-term debt; and *Equity Issuance* is defined as the issuance of equity. All variables are scaled by the lagged total assets.

Figure 1. Financing decisions around the refinement.

Read:

- Figure 1 plots cumulative financing decisions before and after the refinement. Before the event, upgraded and downgraded firms look fairly similar. Afterward, upgraded firms issue more total debt and long-term debt, while downgraded firms issue less.
- The net debt graphs show even sharper divergence because downgraded firms tend to accumulate cash while upgraded firms do not.
- Figure 2 shows that upgraded firms increase capital expenditures more than downgraded firms and accumulate less cash after the refinement.

Economic message: The figures provide the visual version of the paper's main story. A favorable information shock improves debt-market access and appears to turn into higher investment rather than mere balance-sheet reshuffling.

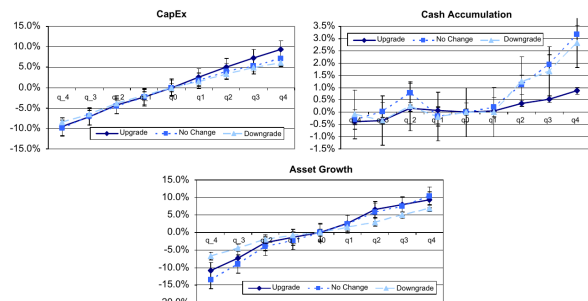


Fig. 2. Effects of rating refinement changes on firm's investment decisions. The graphs show firms' cumulative investment decisions in the four quarters pre- and post-Moody's 1982 rating refinement (denoted as quarter "0" in the graphs). All graphs are re-centered at quarter 0. *CapEx* is the capital expenditure calculated as the change in property, plant, and equipment plus depreciation; *Cash Accumulation* is defined as the change in cash and cash equivalents; and *Asset Growth* is calculated as the change in total assets. All variables are scaled by lagged total assets.

Figure 2. Investment and cash decisions around the refinement.

7.3 Table 3. Predictability of refinement changes using ex ante yield spreads and Table 4, Panel A. Yield-spread changes by refinement outcome

Table 3

Predictability of refinement changes using yield spreads. Panel A reports the results on the predictability of the direction of Moody's rating refinement changes for the 704 firms identified in Lehman Brother's Fixed Income Database (LIFID) based on the credit information embedded in firms' ex ante yield spreads. Ex ante yield spreads, $YieldSpread_{i,t}$, in percent, are measured as the difference between firms' corporate bond yields and equal-duration Treasury bond yields, one month prior to Moody's April 1982 credit rating refinement. Corporate bond yield is the yield-to-maturity on a bond drawn randomly for a given firm based on the monthly pricing information provided by LIFID. Treasury bond yield is the average yield on Treasury bonds for a given duration. Panel B shows ordered probit regression results using rating refinement changes as dependent variable and firms' ex ante yield spreads as independent variable. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

Panel A						
Pre-refinement ratings	Post-refinement ratings	Rating changes	Number of observations	Yield Spread _{<i>i,t</i>}	Standard error	Statistically different from upgrade
Aa	Aa1 or higher	Upgrade	26	0.95	0.15	–
	Aa2	Nochange	46	0.94	0.12	No
	Aa3 or lower	Downgrade	47	1.29	0.13	No
A	A1 or higher	Upgrade	91	1.73	0.17	–
	A2	Nochange	96	1.51	0.12	No
	A3 or lower	Downgrade	81	1.67	0.12	No
Baa	Baa1 or higher	Upgrade	95	2.07	0.12	–
	Baa2	Nochange	31	2.60	0.18	Yes
	Baa3 or lower	Downgrade	58	3.21	0.21	Yes
Ba	Ba1 or higher	Upgrade	41	2.69	0.22	–
	Ba2	Nochange	27	3.89	0.21	Yes
	Ba3 or lower	Downgrade	65	6.00	0.52	Yes
Total	Upgrade		253	1.94	0.09	–
	Nochange		200	1.87	0.10	No
	Downgrade		251	3.08	0.19	Yes
Panel B						
Ordered probit model with refinement rating changes as dependent variables						
Rating class	Aa	A	Baa	Ba		
YieldSpread _{<i>i,t</i>}	0.232 [*] (0.120)	–0.019 (0.053)	0.388 ^{***} (0.078)	0.590 ^{***} (0.085)		
With 1% increase in YieldSpread _{<i>i,t</i>}						
Prob (Upgrade)	–0.061	0.004	–0.232	–0.145		
Prob (NoChange)	–0.030	0.000	0.021	–0.066		
Prob (Downgrade)	0.091	–0.003	0.211	0.211		
Pseudo R ²	0.014	0.000	0.076	0.250		
Number of observations	119	268	184	133		
Number predictable	54	98	115	92		

Table 4

Effects of rating refinement changes on yield spreads. Panel A presents the summary statistics on the change in firms' yield spreads (based on equal-duration Treasury bonds) subsequent to Moody's rating refinements. $YS_{i,t}$ and $YS_{i,t-1}$ denote firms' yield spreads, in percent, in the month of March and April, respectively. ΔYS is defined as $YS_{i,t} - YS_{i,t-1}$. Panel B reports regression results of the subsequent change in firms' yield spreads, both in levels ($YS_{i,t}$) and relative percentage term ($\Delta YS/YS_{i,t-1}$). Upgrade and Nochange are dummy variables representing rating upgrades and no changes resulting from the rating refinement, respectively. A, Baa, and Ba are dummy variables indicating firms' rating classes. Panel C presents summary results of changes in gross bond yields as well as changes in yield spreads calculated using four different measures including measuring yield spreads off equal-duration Treasury bonds, equal-time-to-maturity Treasury bonds, median bond yields of corresponding rating classes, and as residual bond yields. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

Panel A				
Pre-refinement ratings	Rating changes	Average $YS_{i,t-1}$	ΔYS	$\Delta YS/YS_{i,t-1}$
Aa	Upgrade	0.95	-0.08	-0.05
	Nochange	0.91	0.00	0.00
	Downgrade	0.94	0.29	0.48
A	Upgrade	1.73	-0.14	-0.08
	Nochange	1.51	0.14	0.09
	Downgrade	1.67	-0.03	-0.18
Baa	Upgrade	2.07	0.04	0.02
	Nochange	1.71	-0.31	-0.88
	Downgrade	3.21	0.07	0.02
Ba	Upgrade	1.51	0.11	0.07
	Nochange	1.87	0.36	0.19
	Downgrade	6.00	-0.02	-0.15
Total	Upgrade	1.94	-0.02	-0.01
	Nochange	1.87	0.04	0.02
	Downgrade	3.08	0.07	0.02
Baa	Upgrade	2.00	0.09	0.05
	Nochange	2.60	-0.05	-0.19
	Downgrade	6.22	0.05	0.01
Ba	Upgrade	2.69	0.07	0.03
	Nochange	3.89	0.45	0.12
	Downgrade	6.00	-0.02	-0.03
Total	Upgrade	1.94	-0.07	-0.04
	Nochange	1.87	0.12	0.06
	Downgrade	3.08	0.11	0.04

Table 3. Predictability of refinement changes using ex ante yield spreads.

Table 4, Panel A. Yield-spread changes by refinement outcome.

Read:

- Table 3 shows that ex ante yield spreads partly predict the direction of the refinement, especially in lower rating classes such as Baa and Ba.
- This matters because the paper is not claiming that the refinement was pure surprise at the firm-type level. Instead, some information was already embedded in bond yields.
- Table 4, Panel A shows that even after this partial predictability, yield spreads move in the expected direction after the refinement.

Economic message: The market was not completely ignorant before Moody's acted, but the refined rating still added useful public information. This supports the interpretation that the ratings refinement reduced residual information asymmetry.

7.4 Table 4, Panels B–C. Regression evidence on yield spreads and Table 5. Credit-score calculation of the leverage offset

Panel B	AY5					AY5, % _{t-1}				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Upgrade	-0.194*** (0.060)	-0.192*** (0.067)	-0.136*** (0.067)	-0.011*** (0.004)	-0.011*** (0.004)	-0.011*** (0.004)	-0.011*** (0.004)	-0.011*** (0.004)	-0.011*** (0.004)	-0.011*** (0.004)
NoChange	-0.007 (0.070)	-0.007 (0.071)	-0.007 (0.070)	-0.007 (0.004)	-0.007 (0.004)	-0.007 (0.004)	-0.007 (0.004)	-0.007 (0.004)	-0.007 (0.004)	-0.007 (0.004)
AY5 _{t-1}										
AY5 _{t-1} yield _{t-2}										
AY5 _{t-1}		0.026 (0.002)	0.006 (0.001)		0.003 (0.002)	0.003 (0.002)		0.003 (0.002)	0.003 (0.002)	0.003 (0.002)
AY5 _{t-1}		0.015 (0.000)	-0.017 (0.000)		0.002 (0.002)	0.002 (0.002)		0.002 (0.002)	0.002 (0.002)	0.002 (0.002)
AY5 _{t-1}		0.000 (0.000)	-0.009 (0.000)		0.001 (0.002)	0.001 (0.002)		0.001 (0.002)	0.001 (0.002)	0.001 (0.002)
Constant	0.128*** (0.047)	0.099 (0.077)	0.080 (0.094)	0.007*** (0.002)	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)
Number of observations	704	704	704	704	704	704	704	704	704	704
R ²	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02

Panel C	A5yield					A5yieldspread				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Upgrade	-0.154*** (0.058)	-0.194*** (0.059)	-0.192*** (0.060)	-0.192*** (0.061)	-0.192*** (0.062)	-0.192*** (0.063)	-0.192*** (0.064)	-0.192*** (0.065)	-0.192*** (0.066)	-0.192*** (0.067)
NoChange	-0.001 (0.062)	-0.001 (0.063)	-0.001 (0.064)	-0.001 (0.065)	-0.001 (0.066)	-0.001 (0.067)	-0.001 (0.068)	-0.001 (0.069)	-0.001 (0.070)	-0.001 (0.071)
AY5 _{t-1}										
AY5 _{t-1}		0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
AY5 _{t-1}		0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Constant	-0.253*** (0.041)	-0.117*** (0.041)	0.128*** (0.047)	0.107*** (0.050)	0.107*** (0.051)	0.107*** (0.052)	0.107*** (0.053)	0.107*** (0.054)	0.107*** (0.055)	0.107*** (0.056)
Number of observations	704	704	704	704	704	704	704	704	704	704
R ²	0.01	0.02	0.02	0.02	0.02	0.02	0.01	0.02	0.01	0.01

Table 4, Panels B–C. Regression evidence on yield spreads.

Read:

- Table 4 shows that an upgrade lowers yield spreads by about 20 basis points relative to a downgrade. The result is robust to different yield-spread definitions.
- Table 5 asks whether the observed effect understates the true information effect because upgraded firms later issue more debt, which mechanically raises default risk and partially offsets the spread decline.
- Using Altman Z-score and Ohlson O-score calculations, the paper argues that the true rating-refinement effect could be larger than the observed spread change.

Economic message: The cost-of-borrowing effect is economically important. It is not just a mechanical relabeling of rating categories; the market prices the additional information.

Table 5
Estimating the true effect of rating refinement changes using credit score models. Panel A summarizes the original coefficients used in the two credit score models: WC/TA is working capital (data170) divided by total assets (data46); RE/TA is retained earnings (data36) divided by total assets; EBIT/TA is earnings before interest and taxes (data13) divided by total assets; VE/TA is market value of equity (data25 - data199) divided by total debt (data9+data34); S/TA is sales (data12) divided by total assets; Size is the natural log of total assets (adjusted for gross domestic product price level index); TL/TA is total liabilities (data181) divided by total assets; CL/CA is current liabilities (data25) divided by current assets (data44); NI/TA is net income (data172) divided by total assets; FFO/TL is pre-tax income (data170) plus depreciation and amortization (data14) divided by total liabilities; INTWO is an indicator variable equal to one if cumulative net income over the previous two years is negative, and zero otherwise; OENEG is an indicator variable equal to one if owners' equity is negative and zero otherwise; and CHN is the scaled change in net income and is calculated as $(NI_t - NI_{t-1}) / (NI_t + NI_{t-1})$. Regarding the original coefficients in the Altman (1968) firms' default probability is decreasing with the Z-Score. Consistent with Hillegent, Keating, Cram, and Lundstedt (2004), to make it comparable to Ohlson's O-Score, the signs of the original coefficients have been changed and all of the coefficients except S/TA have been multiplied by one hundred, so the Z-Score is increasing in the default probability. See Altman (1968) and Ohlson (1980). Panel B illustrates the marginal impact of an increase in firms' long-term debt issuance on firms' default probabilities and yield spreads. The effect is examined for both the average firm in Moody's 1982 credit rating refinement and the average firm in the original studies. The credit score for the former is calculated based on the annual Compustat data one year prior to the rating refinement. DP₀ is the default probability calculated by converting credit scores through a logistic cumulative distribution function, and YS is the yield spread calculated through a linear approximation based on the change in firms' default probabilities. The average non-bankrupt firms from Altman and Ohlson studies are from 1946 to 1965 and from 1970 to 1976, respectively. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

Panel A										
Altman (1968) model	WC/TA	RE/TA	EBIT/TA	VE/TA	S/TA					
Original coefficients	-1.20	-1.40	-3.30	-0.60	-0.999					
Ohlson (1980) model	Size	TL/TA	WC/TA	CL/CA	NI/TA	FFO/TL	INTWO	OENEG	CHN	Constant
Original coefficients	-0.41***	6.03***	-1.43**	0.08	-2.37**	-1.83***	0.285	-1.72**	-0.52**	v.32

Panel B										
Credit score model	Benchmark	Long-term debt issuance increased by 25% TA				Long-term debt issuance increased by 55% TA				
	DP ₀	DP ₁	ADP ₁	ADP ₀ /DP ₀	~ΔYS	DP ₂	ADP ₂	ADP ₀ /DP ₀	~ΔYS	
Average firm in Moody's 1982 rating refinement										
Z-Score	0.033	0.037	0.004	11.4%	21.3	0.042	0.009	26.8%	50.0	
O-Score	0.028	0.032	0.004	12.4%	23.2	0.039	0.011	37.2%	69.6	
Average non-bankrupt firm in the original studies										
Z-Score	0.008	0.009	0.001	14.3%	26.8	0.010	0.003	34.3%	64.1	
O-Score	0.008	0.009	0.001	12.7%	23.8	0.011	0.003	34.8%	65.1	

Table 5. Credit-score calculation of the leverage offset.

7.5 Table 6, Panels A–B. Long-term and net long-term debt issuance and Table 6, Panels C–F. Short-term, total, and net total debt issuance

Table 6

Effects of rating refinement changes on firm's subsequent debt issuance. Panels A, B, C, D, E, and F present the effects of rating refinement on the change in firms' long-term debt issuance, net long-term debt issuance (long-term debt issuance net of the change in cash and cash equivalents), short-term debt issuance, the ratio of short-term to long-term debt issuance, total debt issuance, and net total debt issuance (total debt issuance net of the change in cash and cash equivalents), respectively. Panels C, D, E, and F have the same regression specification as those of Panels A and B, but report only the results on key variables of interests. All debt issuance are scaled by the lagged total assets. *Upgrade* and *NoChange* are rating upgrades and no changes resulting from Moody's rating refinements, respectively. *YieldSpread* is the firms' ex ante yield spread based off equal-duration Treasury yield; *Size* is calculated as the log of the market value of total assets; *MTBV* is the market-to-book ratio; *Profitability* is the operating income before depreciation and scaled by total assets; *Tangibility* is firms' net property, plant, and equipment scaled by the total assets; and *A₁*, *A₂*, and *A₃* are dummy variables indicating firms' original rating assignments. All accounting variables are yearly results calculated based on quarterly Compustat data. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

Panel A						
	Dependent variable: $\Delta \text{Long-Term DebtIss}_{i,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Upgrade_{it}</i>	0.024** (0.012)	0.021* (0.012)	0.021* (0.011)	0.022* (0.013)	0.024* (0.013)	0.023** (0.011)
<i>NoChange_{it}</i>	0.012 (0.013)	0.018 (0.012)	0.023** (0.011)	0.019 (0.013)	0.021 (0.013)	0.025** (0.011)
<i>YieldSpread_{i,t-1}</i>				-0.002 (0.003)	0.003 (0.004)	0.003 (0.004)
$\Delta \text{Size}_{i,t-1}$		-0.187*** (0.037)	-0.006 (0.042)		-0.189*** (0.037)	-0.008 (0.042)
$\Delta \text{MTBV}_{i,t-1}$		0.147** (0.027)	0.031 (0.029)	0.150** (0.027)	0.034 (0.030)	0.151 (0.030)
$\Delta \text{Profitability}_{i,t-1}$		0.454** (0.164)	0.150 (0.156)	0.454** (0.164)	0.151 (0.156)	0.454** (0.156)
$\Delta \text{Tangibility}_{i,t-1}$		-0.317** (0.126)	-0.266* (0.115)	-0.315** (0.126)	-0.265* (0.115)	-0.315** (0.115)
$\Delta \text{Long-Term DebtIss}_{i,t-1}$			-0.409** (0.057)		-0.409** (0.057)	
Constant	-0.027*** (0.009)	0.005 (0.013)	-0.009 (0.012)	-0.021 (0.013)	0.000 (0.015)	-0.013 (0.013)
Rating class fixed effects		Yes	Yes	Yes	Yes	Yes
Number of observations	266	266	266	266	266	266
R ²	0.02	0.19	0.33	0.02	0.19	0.33

Panel B						
	Dependent variable: $\Delta \text{Net Long-Term DebtIss}_{i,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Upgrade_{it}</i>	0.051*** (0.014)	0.044*** (0.013)	0.046*** (0.013)	0.042*** (0.014)	0.045*** (0.014)	0.047*** (0.013)
<i>NoChange_{it}</i>	0.022 (0.015)	0.028* (0.014)	0.032* (0.013)	0.014 (0.015)	0.028* (0.014)	0.033* (0.013)
<i>YieldSpread_{i,t-1}</i>				-0.008* (0.004)	0.001 (0.005)	0.001 (0.005)
$\Delta \text{Size}_{i,t-1}$		-0.181*** (0.041)	-0.066 (0.044)		-0.182*** (0.041)	-0.066 (0.044)
$\Delta \text{MTBV}_{i,t-1}$		0.145** (0.030)	0.068* (0.032)	0.146** (0.031)	0.069* (0.032)	0.145** (0.032)
$\Delta \text{Profitability}_{i,t-1}$		0.371** (0.184)	0.115 (0.180)	0.371** (0.184)	0.116 (0.180)	0.371** (0.180)
$\Delta \text{Tangibility}_{i,t-1}$		-0.466*** (0.141)	-0.468*** (0.138)	-0.466*** (0.141)	-0.468*** (0.138)	-0.466*** (0.138)
$\Delta \text{Net Long-Term DebtIss}_{i,t-1}$			-0.301*** (0.054)		-0.300*** (0.054)	
Constant	-0.057*** (0.010)	-0.008 (0.015)	-0.02 (0.014)	-0.035** (0.015)	-0.01 (0.016)	-0.022 (0.016)
Rating class fixed effects		Yes	Yes	Yes	Yes	Yes
Number of observations	266	266	266	266	266	266
R ²	0.05	0.25	0.33	0.06	0.25	0.33

Table 6, Panels A–B. Long-term and net long-term debt issuance.

Read:

- Panel A shows that upgraded firms issue around 2 percent of lagged assets more long-term debt than downgraded firms.
- Panel B shows an even larger effect for net long-term debt issuance, around 4–5 percent of lagged assets.
- Panels C–F show weaker short-term debt effects but positive net total debt effects, meaning the financing response is concentrated in long-term debt and net borrowing.

Economic message: Better information does not merely reduce yields in secondary bond markets. It changes firms' financing behavior, especially their ability and willingness to issue long-term debt.

Table 6 (continued)						
Panel A						
	Dependent variable: $\Delta \text{Long-Term DebtIss}_{i,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel C						
	Dependent variable: $\Delta \text{Short-Term DebtIss}_{i,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Upgrade_{it}</i>	-0.001 (0.008)	-0.003 (0.008)	-0.009 (0.007)	-0.006 (0.008)	-0.006 (0.009)	-0.010 (0.007)
<i>NoChange_{it}</i>	-0.007 (0.008)	-0.007 (0.008)	-0.008 (0.007)	-0.011 (0.008)	-0.009 (0.009)	-0.008 (0.007)
<i>YieldSpread_{i,t-1}</i>				-0.004* (0.002)	-0.002 (0.003)	-0.001 (0.003)
R ²	0	0.03	0.29	0.02	0.04	0.29
Panel D						
	Dependent variable: $\Delta \text{Short-Term DebtIss}/\text{Long-Term DebtIss}_{i,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Upgrade_{it}</i>	-44.473* (26.935)	-41.564 (28.916)	-49.009* (27.703)	-46.326 (28.357)	-51.911* (30.857)	-56.545* (29.557)
<i>NoChange_{it}</i>	-19.846 (28.936)	-18.442 (28.883)	-18.209 (28.587)	-21.385 (29.879)	-23.766 (30.842)	-23.587 (29.531)
<i>YieldSpread_{i,t-1}</i>				-1.617 (7.605)	-10.15 (10.553)	-7.451 (10.118)
R ²	0.01	0.03	0.11	0.01	0.03	0.11
Panel E						
	Dependent variable: $\Delta \text{Total DebtIss}_{i,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Upgrade_{it}</i>	0.024* (0.013)	0.018 (0.013)	0.015 (0.013)	0.017 (0.014)	0.019 (0.014)	0.016 (0.013)
<i>NoChange_{it}</i>	0.008 (0.014)	0.014 (0.013)	0.018 (0.013)	0.002 (0.015)	0.015 (0.014)	0.019 (0.013)
<i>YieldSpread_{i,t-1}</i>				-0.006 (0.004)	0.001 (0.005)	0.001 (0.005)
R ²	0.01	0.21	0.25	0.02	0.21	0.25
Panel F						
	Dependent variable: $\Delta \text{Net Total DebtIss}_{i,t+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Upgrade_{it}</i>	0.040*** (0.015)	0.040*** (0.014)	0.039*** (0.014)	0.036*** (0.015)	0.039*** (0.015)	0.039*** (0.015)
<i>NoChange_{it}</i>	0.018 (0.016)	0.023 (0.014)	0.027 (0.014)	0.007 (0.016)	0.023 (0.015)	0.027 (0.015)
<i>YieldSpread_{i,t-1}</i>				-0.012*** (0.004)	-0.001 (0.005)	0.000 (0.005)
R ²	0.04	0.27	0.30	0.07	0.27	0.30

Table 6, Panels C–F. Short-term, total, and net total debt issuance.

7.6 Table 7. Equity issuance after the refinement and Table 8. Overall capital-market access and debt-equity mix

Table 7
Effects of rating refinement changes on firms' subsequent equity issuance. $Equity_{it}$ is defined as firms' equity issuance scaled by the lagged total assets; $Upgrade$ and $NoChange$ are rating upgrades and no changes resulting from Moody's rating refinements, respectively; $YieldSpread_{it}$ is the firms' ex ante yield spread based off equal-duration Treasury yield; $Size$ is calculated as the log of the market value of total assets; MBV is the market-to-book ratio; $Profitability$ is the operating income before depreciation and scaled by total assets; $Tangibility$ is firms' net property, plant, and equipment scaled by the total assets; and A , Baa , and Bu are dummy variables indicating firms' original rating assignments. All accounting variables are yearly results calculated based on quarterly Compustat data. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

	Dependent variable: $\Delta Equity_{it+1}$					
	(1)	(2)	(3)	(4)	(5)	(6)
$Upgrade_{it}$	-0.009 (0.007)	-0.013* (0.007)	-0.012* (0.007)	-0.006 (0.008)	-0.016** (0.008)	-0.016** (0.008)
$NoChange_{it}$	0.005 (0.008)	0.008 (0.008)	0.007 (0.008)	0.008 (0.008)	0.005 (0.008)	0.005 (0.008)
$YieldSpread_{it-1}$				0.003 (0.002)	-0.004 (0.003)	-0.004 (0.003)
$\Delta Size_{it-1}$		0.01 (0.023)	0.015 (0.023)	0.013 (0.023)	0.017 (0.023)	0.017 (0.023)
$\Delta MV/BV_{it-1}$		0.028 (0.017)	0.032 (0.017)	0.025 (0.017)	0.028 (0.017)	0.028 (0.017)
$\Delta Profitability_{it-1}$		-0.043 (0.103)	-0.047 (0.103)	-0.044 (0.103)	-0.048 (0.103)	-0.048 (0.103)
$\Delta Tangibility_{it-1}$		-0.162* (0.079)	-0.154* (0.079)	-0.164* (0.079)	-0.156* (0.079)	-0.156* (0.079)
$\Delta Equity_{it-1}$			-0.073 (0.043)		-0.069 (0.043)	-0.069 (0.043)
Constant	0.009* (0.005)	0.003 (0.008)	0.004 (0.008)	0.000 (0.008)	0.009 (0.009)	0.009 (0.009)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	266	266	266	266	266	266
R^2	0.01	0.12	0.13	0.02	0.13	0.14

Table 7. Equity issuance after the refinement.

Table 8
Effects of rating refinement changes on firms' overall capital market access. $\Delta NetTotalDebt_{it}$ is defined as the change in firms' total debt issuance net of the change in cash and scaled by the lagged total assets; $\Delta Equity_{it}$ is defined as the change in firms' equity issuance and scaled by the lagged total assets; $Upgrade$ and $NoChange$ are rating upgrades and no changes resulting from Moody's rating refinements, respectively; $YieldSpread$ is the firms' ex ante yield spread based off equal-duration Treasury yield; $Size$ is calculated as the log of the market value of total assets; MBV is the market-to-book ratio; $Profitability$ is the operating income before depreciation and scaled by total assets; $Tangibility$ is firms' net property, plant, and equipment scaled by the total assets; and A , Baa , and Bu are dummy variables indicating firms' original rating assignments. All accounting variables are yearly results calculated based on quarterly Compustat data. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

Dependent variable	$\Delta Net Total Debt_{it+1} + \Delta Equity_{it+1}$				$\Delta Net Total Debt_{it+1} - \Delta Equity_{it+1}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$Upgrade_{it}$	0.037** (0.016)	0.025* (0.015)	0.027* (0.017)	0.021 (0.016)	0.059*** (0.017)	0.053*** (0.017)	0.043*** (0.018)	0.057*** (0.018)
$NoChange_{it}$	0.024 (0.017)	0.035* (0.015)	0.015 (0.018)	0.032 (0.016)	0.013 (0.019)	0.017 (0.017)	-0.001 (0.019)	0.019 (0.018)
$YieldSpread_{it-1}$				-0.009 (0.005)	-0.004 (0.005)	-0.004 (0.005)	-0.015* (0.005)	0.004 (0.005)
$\Delta Size_{it-1}$			-0.050 (0.051)	-0.049 (0.051)		-0.189*** (0.055)		-0.192*** (0.056)
$\Delta MV/BV_{it-1}$			0.125* (0.035)	0.126* (0.035)		0.119* (0.042)		0.123* (0.042)
$\Delta Profitability_{it-1}$			0.249 (0.209)	0.251 (0.210)		0.552 (0.240)		0.553 (0.240)
$\Delta Tangibility_{it-1}$			-0.675*** (0.160)	-0.679*** (0.160)		-0.447*** (0.182)		-0.445*** (0.182)
$\Delta Net Total Debt_{it-1} + \Delta Equity_{it-1}$			-0.240* (0.048)	-0.238* (0.048)				
$\Delta Net Total Debt_{it-1} - \Delta Equity_{it-1}$						-0.046 (0.052)		-0.046 (0.052)
Constant	-0.052*** (0.012)	-0.004 (0.016)	-0.027 (0.018)	0.002 (0.018)	-0.069*** (0.013)	-0.005 (0.019)	-0.026 (0.019)	-0.011 (0.021)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	266	266	266	266	266	266	266	266
R^2	0.02	0.32	0.03	0.32	0.05	0.22	0.08	0.22

Table 8. Overall capital-market access and debt-equity mix.

Read:

- Table 7 shows that upgraded firms issue less equity than downgraded firms after the refinement, with the effect around 1–2 percent of lagged assets in several specifications.
- Table 8 combines net debt issuance and equity issuance. Upgraded firms rely more heavily on debt financing relative to equity financing.
- The pattern is consistent with a localized information shock about credit risk: better credit information directly improves debt-market access, not necessarily equity-market access.

Economic message: The refinement shifts the financing mix. Firms that are revealed to be better credit risks substitute toward debt, which supports the view that the rating refinement works through a credit-supply channel.

7.7 Table 9. Investment, cash, and asset-growth responses and Table 10, Panels A–B. Within-rating-class refinements

Table 9

Changes in firm's investment policies following the rating refinement. $\Delta CapEx$ is firms' capital expenditure, which is calculated as the change in firms' net property, plant, and equipment plus depreciation and scaled by lagged total assets; $\Delta CashAccumulation$ is firms' cash accumulation, which is defined as the change in cash and cash equivalents, and scaled by lagged total assets; $\Delta AssetGrowth$ is firms' asset growth rate, which is defined as the change in firms' total assets and scaled by the lagged total assets. All accounting variables are yearly results calculated based on quarterly Compustat data. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

Dependent variable	$\Delta CapEx_{t,t+1}$			$\Delta CashAccumulation_{t,t+1}$			$\Delta AssetGrowth_{t,t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Upgrade_{it}</i>	0.050*** (0.016)	0.033* (0.017)	0.022* (0.013)	-0.021*** (0.007)	-0.020*** (0.008)	-0.018** (0.008)	0.055** (0.025)	0.036 (0.027)	0.029 (0.023)
<i>NoChange_{it}</i>	0.003 (0.018)	-0.010 (0.017)	-0.007 (0.013)	-0.011 (0.008)	-0.009 (0.008)	-0.01 (0.008)	0.033 (0.027)	0.019 (0.024)	0.025 (0.024)
<i>YieldSpread_{t-1}</i>			-0.014** (0.006)	-0.006 (0.005)	0.000 (0.003)	0.001 (0.003)	-0.019* (0.009)	-0.006 (0.008)	
$\Delta Size_{i,t-1}$			-0.325*** (0.042)	-0.010 (0.023)	-0.010 (0.023)	-0.010 (0.023)	-0.506*** (0.080)		
$\Delta MV/BV_{i,t-1}$			0.334*** (0.029)	0.013 (0.018)	0.013 (0.018)	0.013 (0.018)	0.512*** (0.053)		
$\Delta Profitability_{i,t-1}$			-0.046 (0.175)	-0.027 (0.175)	-0.027 (0.175)	-0.027 (0.175)	0.327 (0.307)		
$\Delta Tangibility_{i,t-1}$			-0.777*** (0.124)	-0.189*** (0.065)	-0.302*** (0.072)	-0.302*** (0.072)	-0.199*** (0.199)		
$\Delta CapEx_{i,t-1}$	-0.515*** (0.074)	-0.570*** (0.072)	-0.237*** (0.065)						
$\Delta CashAccumulation_{i,t-1}$				-0.397*** (0.047)	-0.389*** (0.047)	-0.345*** (0.050)			
$\Delta AssetGrowth_{i,t-1}$						-0.353*** (0.068)	-0.369*** (0.067)	-0.186*** (0.069)	
Constant	-0.070*** (0.012)	-0.021 (0.020)	0.023 (0.016)	0.020*** (0.005)	0.013 (0.009)	0.011 (0.009)	-0.087*** (0.019)	-0.027 (0.012)	0.026 (0.027)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	266	266	266	266	266	266	266	266	266
R ²	0.18	0.25	0.57	0.24	0.25	0.27	0.10	0.13	0.39

Table 9. Investment, cash, and asset-growth responses.

Read:

- Table 9 shows that upgraded firms subsequently invest more, save less cash, and grow assets faster than downgraded firms.
- The investment effect is about 3 percent of lagged assets in the controlled specifications.
- Table 10 restricts attention to within-rating-class refinements, addressing the concern that cross-class refinements might reflect broader rating-scale realignment or fundamentals.
- The debt-issuance and cost-of-borrowing results remain strong in the within-class sample.

Economic message: The main result is a real effect, not only a capital-structure effect. When better credit information lowers frictions, firms invest more and hold less cash. The within-rating-class test makes the evidence cleaner by focusing on firms whose broad letter rating did not mechanically change.

Table 10

Effects of within rating class refinement changes on firms' subsequent financing and investment decisions. Panel A presents summary results on firms' cost of borrowing after the rating refinement. Panels B and C present changes in firms' financing decisions and investment policies following the refinement. In Panels B and C, $\Delta YieldSpread$ is the firm's ex ante yield spread based off equal-duration Treasury yield. All accounting variables are yearly results calculated based on quarterly Compustat data. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

Panel A									
Dependent variable	$\Delta YieldSpread$		$\Delta YieldSpread$		Equal-time-to-maturity Treasury bonds		Median bond yield in the corresponding rating class		Residual bond yields
Method	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Upgrade</i>	-0.142*** (0.073)	-0.135*** (0.071)	-0.201*** (0.083)	-0.199*** (0.084)	-0.262*** (0.105)	-0.262*** (0.106)	-0.158*** (0.071)	-0.135*** (0.071)	-0.137*** (0.071)
<i>NoChange</i>	-0.027 (0.076)	-0.033 (0.075)	-0.045 (0.080)	-0.041 (0.081)	-0.094 (0.103)	-0.094 (0.103)	-0.034 (0.076)	-0.033 (0.076)	-0.042 (0.080)
$\Delta_{i,t-1}$			0.110 (0.079)	0.021 (0.081)			0.139 (0.152)	0.057 (0.079)	0.114 (0.079)
$\Delta_{i,t-1}$			0.005 (0.060)	0.027 (0.073)			0.026 (0.132)	-0.127 (0.060)	0.026 (0.060)
$\Delta_{i,t-1}$			0.136 (0.090)	0.103 (0.103)			0.225 (0.140)	0.108 (0.103)	0.160 (0.090)
Constant	-0.257*** (0.050)	-0.322*** (0.075)	-0.125*** (0.023)	-0.093 (0.086)	0.068 (0.073)	-0.049 (0.110)	0.094* (0.050)	0.079 (0.075)	0.148*** (0.040)
Number of observations	571	571	571	571	533	533	571	571	571
R ²	0.01	0.01	0.01	0.01	0.03	0.02	0.01	0.02	0.01
Panel B									
Dependent variable	$\Delta Long-Term Debt_{i,t+1}$			$\Delta Net Long-Term Debt_{i,t+1}$			$\Delta Equity_{i,t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Upgrade</i>	0.032*** (0.014)	0.032*** (0.013)	0.034*** (0.012)	0.051*** (0.016)	0.051*** (0.015)	0.056*** (0.014)	-0.013 (0.006)	-0.020*** (0.006)	-0.020*** (0.006)
<i>NoChange</i>	0.006 (0.013)	0.006 (0.013)	0.002 (0.012)	0.001 (0.016)	0.001 (0.015)	0.009 (0.014)	0.001 (0.006)	0.000 (0.006)	0.000 (0.006)
$\Delta YieldSpread_{i,t-1}$			-0.005 (0.005)	-0.002 (0.005)	-0.002 (0.005)	-0.001 (0.005)	-0.003 (0.005)	-0.003 (0.005)	-0.003 (0.005)
$\Delta Size_{i,t-1}$			-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)
$\Delta MV/BV_{i,t-1}$			0.011 (0.011)	0.011 (0.011)	0.011 (0.011)	0.011 (0.011)	0.011 (0.011)	0.011 (0.011)	0.011 (0.011)
$\Delta Profitability_{i,t-1}$			-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)
$\Delta Tangibility_{i,t-1}$			-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)
$\Delta CapEx_{i,t-1}$			-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)
$\Delta CashAccumulation_{i,t-1}$			-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)
$\Delta AssetGrowth_{i,t-1}$			-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)	-0.001 (0.005)
Constant	-0.070*** (0.012)	-0.021 (0.020)	0.023 (0.016)	0.020*** (0.005)	0.013 (0.009)	0.011 (0.009)	-0.087*** (0.019)	-0.027 (0.012)	0.026 (0.027)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	266	266	266	266	266	266	266	266	266
R ²	0.18	0.25	0.57	0.24	0.25	0.27	0.10	0.13	0.39

Table 10, Panels A–B. Within-rating-class refinements.

7.8 Table 10, Panel C. Within-rating-class investment outcomes and Table 11. Moody's refinement effects controlling for S&P ratings

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Constant	-0.031*** (0.008)	0.002 (0.015)	-0.012 (0.014)	-0.057*** (0.011)	-0.01 (0.017)	-0.026* (0.016)	0.011* (0.007)	0.013 (0.008)	0.012 (0.010)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	218	218	218	218	218	218	218	218	218
R ²	0.03	0.23	0.34	0.05	0.24	0.37	0.02	0.17	0.17

Dependent variable	$\Delta \text{CapEx}_{i,t+1}$			$\Delta \text{CashAccumulation}_{i,t+1}$			$\Delta \text{AssetGrowth}_{i,t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Upgrade _{i,t}	0.064** (0.019)	0.031* (0.016)	0.025* (0.015)	-0.019* (0.008)	-0.019* (0.008)	-0.018* (0.008)	0.057* (0.020)	0.04 (0.020)	0.038 (0.020)
NoChange _{i,t}	-0.002 (0.009)	-0.005 (0.009)	-0.003 (0.015)	-0.009 (0.008)	-0.009 (0.008)	-0.009 (0.008)	0.036 (0.029)	0.029 (0.029)	0.034 (0.029)
SP Rating _{i,t-1}							0.029 (0.029)	0.025 (0.029)	0.025 (0.029)
YieldSpread _{i,t-1}							-0.027* (0.009)	-0.01 (0.009)	-0.01 (0.009)
$\Delta \text{Size}_{i,t-1}$							-0.469** (0.160)	-0.469** (0.160)	-0.469** (0.160)
$\Delta \text{MV/BV}_{i,t-1}$							0.046 (0.050)	0.046 (0.050)	0.046 (0.050)
$\Delta \text{Profitability}_{i,t-1}$							0.471** (0.160)	0.471** (0.160)	0.471** (0.160)
$\Delta \text{Liquidity}_{i,t-1}$							0.403 (0.160)	0.403 (0.160)	0.403 (0.160)
$\Delta \text{CapEx}_{i,t-1}$							-0.013 (0.103)	-0.013 (0.103)	-0.013 (0.103)
$\Delta \text{CashAccumulation}_{i,t-1}$							-0.462** (0.087)	-0.462** (0.087)	-0.462** (0.087)
$\Delta \text{AssetGrowth}_{i,t-1}$							-0.442** (0.045)	-0.442** (0.045)	-0.442** (0.045)
Constant	-0.062*** (0.014)	-0.017 (0.021)	0.021 (0.017)	0.018* (0.009)	0.012 (0.009)	0.012 (0.009)	-0.345** (0.071)	-0.369** (0.071)	-0.289** (0.071)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	218	218	218	218	218	218	218	218	218
R ²	0.14	0.26	0.54	0.13	0.30	0.36	0.10	0.15	0.17

Table 10, Panel C. Within-rating-class investment outcomes.

Read:

- Table 10, Panel C shows that the within-rating-class sample still produces higher capital expenditures and lower cash accumulation for upgraded firms.
- Table 11 controls for preexisting S&P ratings. This matters because S&P had already introduced finer ratings earlier.
- The results remain strong even conditional on S&P ratings, indicating that Moody's refinement was not simply duplicating information already contained in S&P ratings.

Economic message: The information shock is specific and incremental. Moody's refinement still affects outcomes even when alternative rating information is considered.

7.9 Table 11, continued. S&P benchmark results and Table 12. Financing decisions one year before the refinement

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Upgrade _{i,t}	0.064** (0.019)	0.031* (0.016)	0.025* (0.015)	-0.019* (0.008)	-0.019* (0.008)	-0.018* (0.008)	0.057* (0.020)	0.04 (0.020)	0.038 (0.020)
NoChange _{i,t}	-0.002 (0.009)	-0.005 (0.009)	-0.003 (0.015)	-0.009 (0.008)	-0.009 (0.008)	-0.009 (0.008)	0.036 (0.029)	0.029 (0.029)	0.034 (0.029)
SP Rating _{i,t-1}							0.029 (0.029)	0.025 (0.029)	0.025 (0.029)
YieldSpread _{i,t-1}							-0.027* (0.009)	-0.01 (0.009)	-0.01 (0.009)
$\Delta \text{Size}_{i,t-1}$							-0.469** (0.160)	-0.469** (0.160)	-0.469** (0.160)
$\Delta \text{MV/BV}_{i,t-1}$							0.046 (0.050)	0.046 (0.050)	0.046 (0.050)
$\Delta \text{Profitability}_{i,t-1}$							0.471** (0.160)	0.471** (0.160)	0.471** (0.160)
$\Delta \text{Liquidity}_{i,t-1}$							0.403 (0.160)	0.403 (0.160)	0.403 (0.160)
$\Delta \text{CapEx}_{i,t-1}$							-0.013 (0.103)	-0.013 (0.103)	-0.013 (0.103)
$\Delta \text{CashAccumulation}_{i,t-1}$							-0.462** (0.087)	-0.462** (0.087)	-0.462** (0.087)
$\Delta \text{AssetGrowth}_{i,t-1}$							-0.442** (0.045)	-0.442** (0.045)	-0.442** (0.045)
Constant	-0.062*** (0.014)	-0.017 (0.021)	0.021 (0.017)	0.018* (0.009)	0.012 (0.009)	0.012 (0.009)	-0.345** (0.071)	-0.369** (0.071)	-0.289** (0.071)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	218	218	218	218	218	218	218	218	218
R ²	0.14	0.32	0.61	0.30	0.30	0.35	0.29	0.29	0.37

Table 11, continued. S&P benchmark results.

Table 11
Effects of rating refinement changes on firms under Standard and Poor's (S&P) benchmark. Panel A reports effects of rating refinement changes on firms' debt and equity issuance conditional on S&P ratings. Panel B reports effects of rating refinement changes on firms' capital expenditures, cash accumulation, and asset growth. SP Rating is firms' preexisting S&P ratings decoded with 21 for "AAA" and 1 for "C". All accounting variables are yearly results calculated based on quarterly Compustat data. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; * = significant at 10%.

Dependent variable	$\Delta \text{Long-Term Debt}_{i,t+1}$			$\Delta \text{Equity}_{i,t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)
Upgrade _{i,t}	0.038** (0.015)	0.066** (0.018)	0.071** (0.017)	0.004 (0.020)	-0.009 (0.011)	-0.022* (0.010)
NoChange _{i,t}	0.031* (0.015)	0.051** (0.018)	0.058** (0.017)	0.017 (0.020)	0.019 (0.011)	0.007 (0.010)
SP Rating _{i,t-1}						
YieldSpread _{i,t-1}						
$\Delta \text{Size}_{i,t-1}$						
$\Delta \text{MV/BV}_{i,t-1}$						
$\Delta \text{Profitability}_{i,t-1}$						
$\Delta \text{Liquidity}_{i,t-1}$						
$\Delta \text{CapEx}_{i,t-1}$						
$\Delta \text{CashAccumulation}_{i,t-1}$						
$\Delta \text{AssetGrowth}_{i,t-1}$						
Constant	-0.036 (0.021)	0.105** (0.075)	0.153** (0.069)	0.040** (0.015)	-0.054 (0.045)	-0.059 (0.045)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	181	181	181	181	181	181
R ²	0.04	0.3	0.42	0.07	0.19	0.21

Dependent variable	$\Delta \text{CapEx}_{i,t+1}$			$\Delta \text{CashAccumulation}_{i,t+1}$			$\Delta \text{AssetGrowth}_{i,t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Upgrade _{i,t}	0.042** (0.021)	0.046** (0.022)	0.053** (0.022)	-0.018* (0.009)	-0.018* (0.009)	-0.017* (0.009)	0.071** (0.022)	0.071** (0.022)	0.068** (0.022)
NoChange _{i,t}	-0.019 (0.021)	-0.010 (0.022)	0.004 (0.018)	0.003 (0.009)	0.002 (0.009)	-0.004 (0.009)	0.024 (0.022)	0.023 (0.022)	0.046 (0.020)
SP Rating _{i,t-1}									
YieldSpread _{i,t-1}									
$\Delta \text{Size}_{i,t-1}$									
$\Delta \text{MV/BV}_{i,t-1}$									
$\Delta \text{Profitability}_{i,t-1}$									
$\Delta \text{Liquidity}_{i,t-1}$									
$\Delta \text{CapEx}_{i,t-1}$									
$\Delta \text{CashAccumulation}_{i,t-1}$									
$\Delta \text{AssetGrowth}_{i,t-1}$									
Constant	-0.062*** (0.014)	-0.017 (0.021)	0.021 (0.017)	0.018* (0.009)	0.012 (0.009)	0.012 (0.009)	-0.345** (0.071)	-0.369** (0.071)	-0.289** (0.071)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	218	218	218	218	218	218	218	218	218
R ²	0.14	0.26	0.54	0.13	0.30	0.36	0.10	0.15	0.17

Table 11. Moody's refinement effects controlling for S&P ratings.

Table 12
Cross-section variation in firms' financing decisions one year prior to Moody's rating refinement. Panel A estimates firms' likely rating changes based on their actual rating refinement changes, and Panel B defines firms' likely rating outcomes based on their ex ante yield spreads. All accounting variables are yearly results calculated based on quarterly Compustat data. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; * = significant at 10%.

Dependent variable	Panel A						Panel B					
	$\Delta \text{Long-Term Debt}_{i,t-1}$			$\Delta \text{Equity}_{i,t-1}$			$\Delta \text{Long-Term Debt}_{i,t-1}$			$\Delta \text{Equity}_{i,t-1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(1)	(2)	(3)	(4)	(5)	(6)
Upgrade _{i,t-1}	0.010 (0.013)	0.001 (0.013)	0.002 (0.013)	0.013 (0.009)	0.016* (0.010)	0.006 (0.006)	0.008 (0.013)	0.010 (0.013)	0.011 (0.012)	-0.003 (0.009)	-0.004 (0.010)	-0.002 (0.006)
NoChange _{i,t-1}	0.019 (0.014)	0.013 (0.014)	0.012 (0.013)	-0.004 (0.010)	0.000 (0.009)	0.000 (0.009)	-0.003 (0.013)	-0.005 (0.013)	-0.001 (0.012)	0.008 (0.009)	0.009 (0.010)	0.006 (0.006)
$\Delta \text{Size}_{i,t-1}$												
$\Delta \text{MV/BV}_{i,t-1}$												
$\Delta \text{Profitability}_{i,t-1}$												
$\Delta \text{Liquidity}_{i,t-1}$												
$\Delta \text{Net LT Debt}_{i,t-1}$												
$\Delta \text{Equity}_{i,t-1}$												
Constant	0.000 (0.010)	0.005 (0.016)	-0.013 (0.016)	-0.009 (0.007)	0.005 (0.012)	0.000 (0.007)	0.009 (0.008)	0.007 (0.016)	-0.013 (0.016)	-0.006 (0.006)	0.010 (0.012)	0.001 (0.007)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	252	252	252	252	252	252	252	252	252	252	252	252
R ²	0.01	0.08	0.15	0.01	0.05	0.08	0.00	0.08	0.15	0.00	0.04	0.08

Table 12. Financing decisions one year before the refinement.

Read:

- The continuation of Table 11 completes the S&P benchmark robustness test.
- Table 12 checks whether firms that would later be upgraded were already changing financing before the refinement.
- The pre-refinement differences are weak or absent, which supports the parallel-trends logic behind the difference-in-differences interpretation.

Economic message: If the results were driven only by unobserved firm quality or preexisting investment opportunities, similar financing differences should appear before the refinement. Table 12 shows they do not appear in the same way.

7.10 Table 13. Industry fixed effects and Figure 3. Distribution of financing decisions before and after the refinement

Table 13
Effects of rating refinement changes with industry fixed effect. Panel A reports effects on firms' debt and equity issuance and Panel B reports the effects on firms' capital expenditures, cash accumulation, and asset growth. All accounting variables are yearly results calculated based on quarterly Compustat data. Standard errors are in parentheses. *** = significant at 1%; ** = significant at 5%; and * = significant at 10%.

Dependent variable	$\Delta \text{Long-Term Debt}_{i,t+1}$			$\Delta \text{Net Long-Term Debt}_{i,t+1}$			$\Delta \text{Equity}_{i,t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Upgrade _{it}	0.020 (0.014)	0.024** (0.014)	0.025*** (0.013)	0.035*** (0.016)	0.041*** (0.015)	0.044*** (0.014)	-0.013 (0.009)	-0.019* (0.009)	-0.017* (0.009)
NoChange _{it}	0.007 (0.013)	0.021 (0.013)	0.025* (0.012)	0.016 (0.015)	0.032* (0.014)	0.038* (0.013)	0.006 (0.008)	0.003 (0.008)	0.003 (0.008)
YieldSpread _{it-1}	0.002 (0.004)	0.003 (0.004)	0.003 (0.004)	-0.001 (0.004)	0.002 (0.005)	0.002 (0.005)	0.004 (0.002)	-0.003 (0.003)	-0.003 (0.003)
$\Delta \text{Size}_{i,t-1}$	-0.208** (0.041)	-0.021 (0.046)	-0.021 (0.046)	-0.206** (0.045)	-0.081* (0.047)	-0.081* (0.047)	0.02 (0.026)	0.026 (0.026)	0.01 (0.026)
$\Delta \text{MV/BV}_{i,t-1}$	0.146** (0.031)	0.017 (0.032)	0.017 (0.032)	0.155** (0.034)	0.079* (0.034)	0.079* (0.034)	0.008 (0.019)	0.01 (0.019)	0.01 (0.019)
$\Delta \text{Profitability}_{i,t-1}$	0.441*** (0.167)	0.167 (0.158)	0.167 (0.158)	0.424*** (0.184)	0.178 (0.177)	0.178 (0.177)	-0.078 (0.105)	-0.087 (0.104)	-0.087 (0.104)
$\Delta \text{Tangibility}_{i,t-1}$	-0.294** (0.130)	-0.265** (0.120)	-0.265** (0.120)	-0.648*** (0.143)	-0.448*** (0.138)	-0.448*** (0.138)	-0.123 (0.082)	-0.107 (0.082)	-0.107 (0.082)
$\Delta \text{Long-Term Debt}_{i,t-1}$			-0.400** (0.058)						
$\Delta \text{Net Long-Term Debt}_{i,t-1}$						-0.317*** (0.053)			
$\Delta \text{Equity}_{i,t-1}$									-0.092** (0.044)
Constant	-0.106*** (0.025)	-0.036 (0.029)	-0.036 (0.026)	-0.128*** (0.028)	-0.038 (0.032)	-0.038 (0.030)	-0.029* (0.015)	-0.019 (0.018)	-0.026 (0.018)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	266	266	266	266	266	266	266	266	266
R ²	0.08	0.22	0.34	0.15	0.30	0.39	0.07	0.15	0.17

Dependent variable	$\Delta \text{CapEx}_{i,t+1}$			$\Delta \text{Cash Accumulation}_{i,t+1}$			$\Delta \text{Asset Growth}_{i,t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Upgrade _{it}	0.027 (0.018)	0.018 (0.018)	0.017 (0.015)	-0.016* (0.008)	-0.014 (0.009)	-0.014 (0.009)	0.021 (0.027)	0.008 (0.029)	0.007 (0.025)
NoChange _{it}	-0.011 (0.016)	-0.017 (0.017)	-0.007 (0.014)	-0.013 (0.008)	-0.01 (0.008)	-0.013 (0.008)	0.007 (0.025)	-0.002 (0.026)	0.012 (0.023)
YieldSpread _{it-1}		-0.009 (0.006)	-0.007 (0.005)		0.001 (0.003)	0.001 (0.003)		-0.016 (0.009)	-0.012 (0.008)
$\Delta \text{Size}_{i,t-1}$		-0.342*** (0.046)	-0.342*** (0.046)		0.007 (0.025)	0.007 (0.025)		-0.590*** (0.088)	-0.590*** (0.088)
$\Delta \text{MV/BV}_{i,t-1}$		0.336** (0.033)	0.336** (0.033)		0.004 (0.019)	0.004 (0.019)		0.477*** (0.059)	0.477*** (0.059)
$\Delta \text{Profitability}_{i,t-1}$		-0.038 (0.177)	-0.038 (0.177)		-0.065 (0.105)	-0.065 (0.105)		0.207 (0.302)	0.207 (0.302)
$\Delta \text{Tangibility}_{i,t-1}$		-0.780*** (0.126)	-0.780*** (0.126)		0.165* (0.072)	0.165* (0.072)		-0.25 (0.197)	-0.25 (0.197)
$\Delta \text{CapEx}_{i,t-1}$	-0.546*** (0.070)	-0.571*** (0.070)	-0.571*** (0.067)						
$\Delta \text{Cash Accumulation}_{i,t-1}$				-0.417*** (0.046)	-0.415*** (0.047)	-0.381*** (0.050)			
$\Delta \text{Asset Growth}_{i,t-1}$							-0.389*** (0.064)	-0.390*** (0.065)	-0.142*** (0.070)
Constant	-0.210*** (0.025)	-0.163*** (0.033)	-0.067 (0.031)	0.020* (0.012)	0.016 (0.016)	0.012 (0.018)	-0.288*** (0.039)	-0.243*** (0.052)	-0.057 (0.052)
Rating class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	266	266	266	266	266	266	266	266	266
R ²	0.33	0.35	0.59	0.31	0.32	0.33	0.26	0.27	0.45

Table 13. Industry fixed effects.

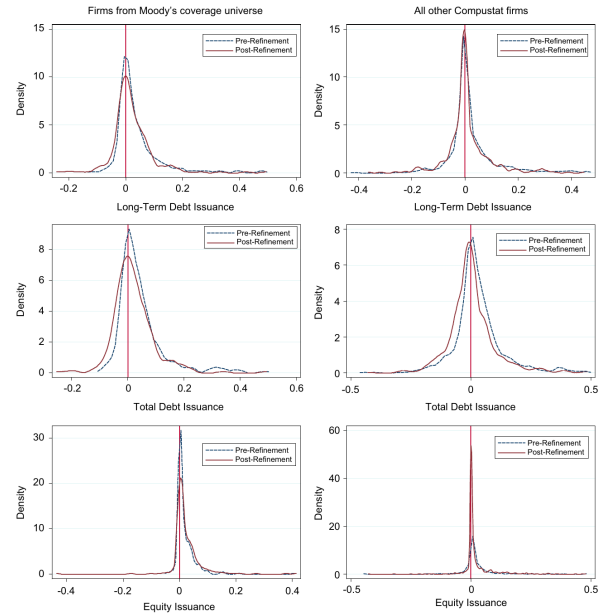


Fig. 3. Probability density graphs of firms' financing decisions before and after Moody's rating refinement. Kernel density estimation is performed using the Epanechnikov kernel with optimal bandwidth based on the Silverman formula. The panel on the left shows the kernel density graphs of the debt and equity issuance for firms from Moody's coverage universe only. The panel on the right shows kernel density graphs of the debt and equity issuance for all Compustat firms excluding firms from Moody's coverage universe. All financing variables are scaled by the lagged total assets.

Figure 3. Distribution of financing decisions before and after the refinement.

Read:

- Table 13 adds industry fixed effects. The main financing effects are still present, especially for long-term and net long-term debt issuance.
- Figure 3 compares the distribution of debt and equity issuance before and after the refinement for Moody's covered firms and other Compustat firms.

- Moody’s covered firms show greater post-refinement dispersion in financing decisions, consistent with more precise information separating firms into different credit-quality groups.
- Other Compustat firms do not show the same distributional pattern.

Economic message: The effects are not simply an industry-investment-opportunity story. The distributional evidence is consistent with the idea that finer credit information causes financing decisions to spread out according to revealed firm quality.

8 Identification and interpretation

8.1 Why the refinement can be interpreted causally

The paper’s identification comes from the institutional design of the refinement. Moody’s did not announce firm-level news; it changed the rating format. Because the change was simultaneous and based on already held information, differences across firms in refined ratings can be interpreted as differences in newly revealed credit quality rather than contemporaneous firm shocks.

8.2 Main threats and paper responses

- **Predictability from yields.** The paper shows that ex ante yield spreads partly predict rating changes, but the refinement still affects borrowing costs and firm policies after controlling for these spreads.
- **Other rating information.** S&P had already refined its ratings, but the results survive controls for S&P ratings.
- **Cross-class rating changes.** Results survive when the analysis is restricted to within-rating-class refinements.
- **Industry investment opportunities.** Results survive industry fixed effects.
- **Pre-trends.** Firms do not show the same financing differences one year before the refinement.

8.3 What the paper does not fully prove

The paper is persuasive that information precision matters, but the exact decomposition of the effects is harder to pin down. The observed post-refinement quantity changes may combine a pure cost-of-capital effect, a debt-supply effect, and a timing acceleration effect for projects previously postponed under information asymmetry.

9 Short summary

Tang uses Moody’s 1982 credit rating format refinement to study whether credit-market information asymmetry affects firms’ financing and investment. The refinement split broad rating classes into finer categories, revealing more information about credit quality without corresponding firm-specific changes in fundamentals.

Firms receiving favorable refined ratings experience lower borrowing costs, issue more long-term debt, issue less equity, invest more, save less cash, and grow assets faster than downgraded firms.

The evidence remains robust to controls for prior yield spreads, S&P ratings, within-rating-class refinements, industry fixed effects, and pre-refinement financing behavior.

The main lesson is that information asymmetry in credit markets has real effects. Public certification by a rating agency can change not only prices in the bond market but also corporate financing capacity and investment behavior.

10 Conclusion

The paper provides clean evidence that information production and disclosure by rating agencies matter for real corporate decisions. By making credit quality more transparent, Moody's 1982 refinement reallocated credit-market access: better-quality firms borrowed more cheaply and expanded investment, while lower-quality firms became relatively more constrained and accumulated cash.

The broader implication is that financial intermediation affects the real economy partly through information. Credit ratings can relax or tighten firms' access to external finance, thereby shaping capital structure, investment, and growth.